PH 101: Special Theory of Relativity **Department of Physics**
Tutorial 1 **Indian Institute of Technology Guwahati**

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1. Show that the wave equation

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} - \frac{1}{c^2}\frac{\partial^2 \phi}{\partial t^2} = 0,$$

   is not invariant under Galilean transformation, where c is the velocity of light.

2. A train with proper length L moves at a speed $5c/13$ with respect to the ground. A ball is thrown from the back of the train to front. The speed of the ball with respect to the train is $c/3$. As viewed by someone on the ground, how much time does the ball spend in the air, and how far does it travel?

3. Two events **A** and **B** are described by space-time coordinates in **S** frame by
   A: $x_1 = 10$ km, $y_1 = 0, z_1 = 0, t_1 = 2 \times 10^{-5}$ sec;
   B: $x_2 = 100$ km, $y_2 = 0, z_2 = 0, t_2 = 10^{-5}$ sec.
   Find out if there is an inertial frame $\mathbf{S}'$ (moving with velocity v along the $x = x'$ axis) in which **A** and **B** are simultaneous. If so, find its velocity with respect to **S**.

4. Triplet A stays on the earth. Triplet B travels at speed $4c/5$ to a planet (a distance L away from earth) and back. Triplet C travels out to the planet at speed $3c/4$, and then returns at the necessary speed to arrive back exactly when B does. [Triplet A,B,C: are of same age by birth.]
   (a) How much does each triplet age during this process?
   (b) Who is youngest?

5. A spaceship is moving away from the earth at a speed $v = 0.8c$. When the spaceship is at a distance of 6.66×10^8 km from the earth as measured in earth's reference frame, a radio signal is sent out to the spaceship by an observer on the earth. How long will it take for the signal to reach the spaceship
   (a) as measured in the earth's reference frame.
   (b) as measured in spaceship's reference frame.

6. A train and a tunnel both have proper lengths L. The train speeds toward the tunnel, with speed v. A bomb is located at the front of the train. The bomb is designed to explode when the front of the train passes the far end of the tunnel. A deactivation sensor is located at the back of the train. When the back of the train passes the near end of the tunnel, this sensor *tells* the bomb to disarm itself. Does the bomb explode?